La Scuola d'Italia / Dr. Huson / IB Math (a)
14 March 2026

Student ID:

7.11 PreQuiz: Logarithmic and Exponential Functions (calculator)

1. [Maximum mark: 6]

   Maya invests \$1800 in an account that pays interest of 5 % per annum, compounded annually. No further deposits or withdrawals are made.

   (a) Show that the value of the account at the end of the first year is \$1890. [1]

   (b) Find the value of the account at the end of 8 years, correct to the nearest cent. [2]

   (c) Find the smallest number of full years required for the value of the account to first exceed \$2700. [3]

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2. [Maximum mark: 6]

The number of followers of a school club account is modelled by

$$F(t) = 1200e^{0.18t}$$

where t is the number of years after the account is created.

(a) Find the number of club followers after one year. [1]

(b) Find the number of followers after 4 years, correct to the nearest whole number. [2]

(c) Find the time, in years, when the number of followers first reaches 3000. Give your answer correct to two decimal places. [3]